

RESHAM JOSHI



CONTACT

Phone

+49 176 27800254

E-Mail

reshamjoshi30@yahoo.com

LinkedIn

Resham Joshi

Address

Wollstraße 12,
67659 Kaiserslautern,
Germany

ABOUT ME

I am a Computer Science graduate with 3+ years of experience in machine learning, knowledge graphs, and full-stack software development across research and industry. Skilled in Python, PyTorch, reinforcement learning, and building end-to-end AI systems, from data pipelines to production deployment. Motivated to apply AI and software engineering skills to build reliable, practical systems that solve real problems.

LANGUAGE

English: (Fluent, IELTS 7.5)

Deutsch: (B1, Goethe actively learning B2)

Hindi: (Native)

Marathi: (Native)

SOFT SKILLS

- Initiative & Ownership
- Communication skills
- Adaptive to situation
- Time Management
- Team Collaboration

Hobbies

- Playing chess
- Singing
- Trekking
- Cooking

• KEY SKILLS

Programming: Python, JavaScript, PHP, Java, C, C++

AI & Domain Expertise: Knowledge Graphs, Ontology Engineering (OWL 2 DL), Reinforcement Learning (Double DQN), LLM Integration (GPT-4o mini), Prompt Engineering, Graph Neural Networks, Causal Discovery (DAGMA), NLP

Libraries/Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, React.js, Node.js, Express.js, RDFLib

Database: MongoDB, Neo4j, SQL, SAP Hybris

PROFESSIONAL EXPERIENCE

Student Assistant, TUM School of Computation, Information and Technology, Munich, Germany
10/2025 - Present

- Develop and maintain production modules on Koinon, TUM's live university platform (PHP, jQuery), with CI/CD pipelines, automated testing, code reviews, and staged rollouts.
- Building an LLM-powered chatbot integrated into Koinon to guide users through the platform, helping them navigate and discover features more easily.
- Developed an LLM-based compliance bot that checks developer code against the project's CLAUDE.md rules before deployment, helping team leads confirm changes are production-ready without breaking existing functionality.
- Contributed workflow improvements reducing faculty administrative effort by nearly 50% through software redesign and process automation.

Master's Thesis, Technical University of Kaiserslautern (TUK), Kaiserslautern, Germany
03/2025 - 05/2026

Topic: Knowledge Graph Quality Improvement via Reinforcement Learning

- Built an autonomous AI agent (Double DQN, dueling architecture, PyTorch) with a generative AI pipeline (GPT-4o mini, NLP) to fully automate knowledge graph repair, achieving 88-96% repair success at 5.1 steps/episode on a 2,000+ triple automotive electronics graph.
- Formulated repair as a Markov Decision Process (6-action environment, action masking, shaped rewards, prioritised experience replay) and integrated OWL reasoners (Konclude, Hermit) as semantic validators.
- Implemented LLM-based triple extraction (OpenAI API, prompt engineering, RDFLib) and a Human-in-the-Loop protocol shifting from expert guided to fully autonomous operation.

Research Assistant, Fraunhofer IGC, Augsburg, Germany
12/2023 - 09/2025

- Deployed an autoencoder based anomaly detection system for live manufacturing operations at around 90% accuracy, with automated defect classification and REST API integration.
- Owned the full ML lifecycle: data ingestion, feature engineering, model training (PyTorch), Docker containerisation, deployment, and production monitoring.
- Improved root cause analysis by combining DAGMA causal discovery with Graph Neural Networks, engineered Neo4J knowledge graphs over multi-parameter production data for downstream ML.
- Built a React.js dashboard translating model outputs into actionable insights for non-technical stakeholders.

Application Engineer, Accenture Solutions Pvt. Ltd., Pune, India
08/2021 - 09/2022

- Delivered React.js and Angular features for high-traffic e-commerce platforms, maintained MongoDB and SAP Hybris at 99.9% uptime.
- Shipped MERN-stack features in agile sprint cycles, reducing sprint delays by 10%.

Intern, Cognifront Academy, Nashik, India

07/2020 - 08/2020

- Implemented supervised and unsupervised machine learning algorithms across 15+ problem statements.
- Interacted with customers to identify the most suitable algorithm and deliver the best possible solution.

Intern, Quick Heal and Maharashtra Police, India

06/2020 - 09/2020

- Interacted with over 2,000 students and teachers to raise awareness about cyber security and the rise in cybercrime caused by lack of awareness.

EDUCATION

Master of Science in Computer Science (Intelligent Systems)

RPTU, Kaiserslautern, Germany

10/2022 - 10/2026

Specialization:

- Machine Learning, Deep Learning.
- Natural Language Processing, Reinforcement Learning.
- Knowledge Graphs, Data Science.
- Data Visualization.

Bachelor of Engineering in Computer Engineering

Savitribai Phule Pune University, Pune, India

07/2017 - 05/2021

Specialization:

- Machine Learning, Data Science.
- Fundamentals of Programming Languages (Python).
- Artificial Intelligence & Robotics.
- Project: Staff Leave and Salary Management System (MERN stack, React.js) automated the leave application and salary management process for staff.

EDUCATIONAL PROJECTS

Academic Project, DFKI, Kaiserslautern, Germany

04/2024 - 10/2024

- Built a full-stack real-time visualisation and debugging platform for multi-agent AI systems (React.js, Node.js, Express.js, MongoDB), improving researcher debugging efficiency by 100%.
- Surfaced 8 previously undetected critical issues in agent interactions through structured validation sessions, improving system reliability and reproducibility.

VOLUNTARY AND LEADERSHIP ACTIVITY

- Educoin NGO, Nashik, India - Supported education and healthy living initiatives for underprivileged children.
- Rotaract Club of Pune Mid-East NGO, Pune, India - Organised donation drives and helped recondition old schools in rural areas.

AWARDS AND CERTIFICATION

- Participated in Code Sprint, 10th National Level Tech Symposium.
- Certificate of Code Challenge, Logmieer Carnival 2019.
- Attended workshops on Artificial Intelligence and "Code is Poetry."
- Zensar Technologies Employability Skill Development Programme.
- Quick Heal Cyber Security Awareness Program.